Question 1

Q1.1 **Part A:** Why does dye spread through Corridors & not walls?

→ $\frac{\partial c}{\partial t} = \nabla \cdot (\alpha \nabla c)$ → rate of change of concentration at any point is proportional to α, where α is large the region responds fast and allows large fluxes, where α is small the region is inert

→ $\underline{J} = -\alpha \nabla c$ → magnitude of transport is determined by gradient and α so even with a large gradient if α is tiny the flux is basically 0

$J_{liquid} = -10^{-1} \cdot \nabla c \gg 0$ $\quad$ $J_{wall} = -10^{-8} \cdot \nabla c \simeq 0$

→ Dirichlet Boundary Conditions $C_{in} = 1$ $C_{out} = 0$ give us a concentration difference across the image

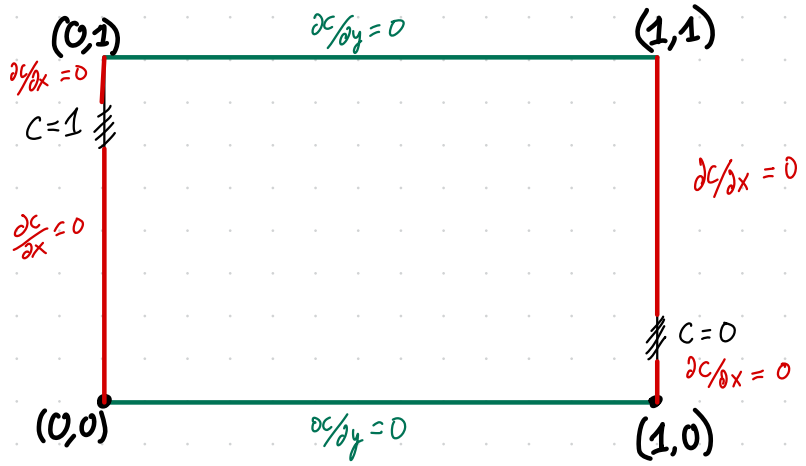
- Same as Lecture 2, 2 fixed temp ends results in a gradient along the conducting path
- Here the gradient will only develop meaningfully along the connected path

Part B: BC's on remaining walls

4 outer walls			Condition
Top edge	$y = 1$	$x\ (0,1)$	$\partial c/\partial y = 0$
Bottom edge	$y = 0$	$x\ (0,1)$	$\partial c/\partial y = 0$
Left edge	$y\ (0 < 0.86,\ 0.94 < 1)$	$x = 0$	$\partial c/\partial x = 0$
Right edge	$y\ (0 < 0.06,\ 0.14 < 1)$	$x = 1$	$\partial c/\partial x = 0$

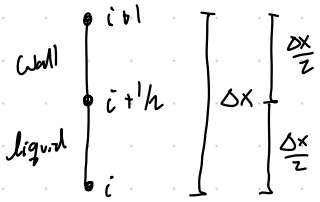
for no mass flux $\underline{J} \cdot \hat{n}$ must be 0

$$\underline{J} \cdot \hat{n} = -\alpha \frac{\partial c}{\partial n} \qquad \alpha \neq 0 \text{ so } \frac{\partial c}{\partial n} \equiv 0$$



Q1.2 Part A One sided diff in liquid only Region

→ physical diffusive flux $J = \alpha \frac{\partial c}{\partial x}$



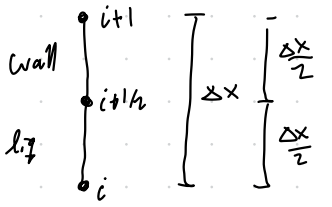
$$\frac{\partial c}{\partial x} \approx \frac{\Delta c}{\Delta x} = \frac{\text{diff in concentration}}{\text{diff in distance}}$$

for liquid only side $(i \to i+1/2)$

$$\left[\frac{\partial c}{\partial x}\Big|_{i+1/2}^{liq} = \frac{C_{i+1/2} - C_i}{\Delta x/2}\right] \times \alpha \Rightarrow J_{i+1/2}^{liq} = \alpha_{liq} \frac{C_{i+1/2} - C_i}{\Delta x/2} = \boxed{\frac{2\alpha_{liq}}{\Delta x}(C_{i+1/2} - C_i) = J_{i+1/2}^{liq}}$$

to find flux

Part B One sided diff in Wall only Region



α_{wall}
$i+1/2 \to i+1$

$$\frac{\partial c}{\partial x}\Big|_{i+1/2}^{wall} \approx \left[\frac{C_{i+1} - C_{i+1/2}}{\Delta x/2}\right] \alpha_{wall} \Rightarrow \boxed{J_{i+1/2}^{wall} = \frac{2\alpha_w}{\Delta x}(C_{i+1} - C_{i+1/2})}$$

for flux

Part C Central diff across wall-liquid interface

$$J_{i+1/2}^{central} = \alpha^* \frac{C_{i+1} - C_i}{\Delta x}$$

$$J^{liq} = J^{wall}$$

$$\frac{\cancel{2}\alpha_{liq}}{\cancel{\Delta x}}(C_{i+1/2} - C_i) = \frac{\cancel{2}\alpha_{wall}}{\cancel{\Delta x}}(C_{i+1} - C_{i+1/2})$$

$$C_{i+1/2} = \frac{\alpha_{wall} C_{i+1} + \alpha_{liq} C_i}{\alpha_{liq} + \alpha_{wall}}$$

$$J_{i+1/2} = \frac{2\alpha_{liq}}{\Delta x}(C_{i+1/2} - C_i) \Rightarrow \frac{2\alpha_{liq}}{\Delta x}\left(\frac{\alpha_w C_{i+1} + \alpha_l C_i}{\alpha_l + \alpha_w} - C_i\right) \Rightarrow \frac{2\alpha_{liq}}{\Delta x} \cdot \frac{\alpha_w C_{i+1} + \alpha_l C_i - (\alpha_l + \alpha_w)C_i}{\alpha_l + \alpha_w}$$

$$\boxed{J_{i+1/2} = \frac{2\alpha_l \alpha_w}{\alpha_l + \alpha_w} \cdot \frac{C_{i+1} - C_i}{\Delta x}}$$

$$\alpha^* = \frac{2\alpha_l \alpha_w}{\alpha_l + \alpha_w} \quad \text{harmonic mean}$$

Part D: Arithmetic avg fail

$$\alpha^*_{arith} = \frac{\alpha_l + \alpha_w}{2} \Rightarrow J^a = \frac{\alpha_l + \alpha_w}{2} \frac{C_{i+1} - C_i}{\Delta x}$$

↳ check with flux continuity

$$\alpha_l (C_{i+1/2} - C_i) = \alpha_w (C_{i+1} - C_{i+1/2})$$

$$\alpha_l \left(\frac{C_{i+1} - C_i}{2}\right) \neq \alpha_w \left(\frac{C_{i+1} - C_i}{2}\right)$$

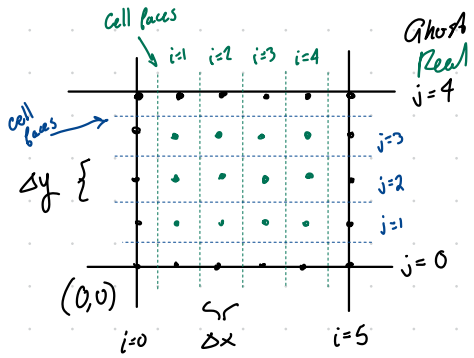
Q1.3 Part A Sketch the grid

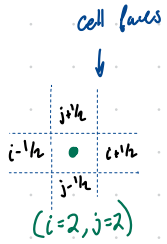
→ Cell Centered: unknowns are in middle of cell not boundaries or edge

→ Ghost Cells: fake cells placed outside of system that allow us to apply BCs

	Real		Ghost
x →	$N_x = 4$	→	$i = 0$ & $i = N_x + 1 = 5$
y →	$N_y = 3$	→	$j = 0$ & $j = N_y + 1 = 4$

→ full grid = 6×5





Space between cells

$$\Delta x = \frac{L_x}{N_x} = \frac{L_x}{4}$$

$$\Delta y = \frac{L_y}{N_y} = \frac{L_y}{3}$$

Part B Derive 2D Spatial Discretization

governing 2D unsteady diffusion

$$\frac{\partial c}{\partial t} = \alpha\left(\frac{\partial^2 c}{\partial x^2} + \frac{\partial^2 c}{\partial y^2}\right)$$

→ discretize RHS spatial terms for an internal cell (i,j)

X-Direction

→ Forward Expansion

$$c_{i+1,j} = c_{i,j} + \Delta x \cancel{\frac{\partial c}{\partial x}\Big|_{i,j}} + \frac{\Delta x^2}{2!}\frac{\partial^2 c}{\partial x^2}\Big|_{i,j} + \cancel{\frac{\Delta x^3}{3!}\frac{\partial^3 c}{\partial x^3}\Big|_{i,j}} + O(\Delta x^4)$$

higher order terms

→ Backward Expansion

$$c_{i-1,j} = c_{i,j} - \Delta x \cancel{\frac{\partial c}{\partial x}\Big|_{i,j}} + \frac{\Delta x^2}{2!}\frac{\partial^2 c}{\partial x^2}\Big|_{i,j} - \cancel{\frac{\Delta x^3}{3!}\frac{\partial^3 c}{\partial x^3}\Big|_{i,j}} + O(\Delta x^4)$$

→ Sum

$$c_{i+1,j} + c_{i-1,j} = 2c_{i,j} + \Delta x^2 \frac{\partial^2 c}{\partial x^2}\Big|_{i,j} + O(\Delta x^4)$$

$$\frac{\partial^2 c}{\partial x^2}\Big|_{i,j} \approx \frac{c_{i+1,j} - 2c_{i,j} + c_{i-1,j}}{\Delta x^2} + O(\Delta x^2)$$

Y-Direction

→ Forward Expansion

$$c_{i,j+1} = c_{i,j} + \Delta y \frac{\partial c}{\partial y}\Big|_{i,j} + \frac{\Delta y^2}{2!}\frac{\partial^2 c}{\partial y^2}\Big|_{i,j} + \frac{\Delta y^3}{3!}\frac{\partial^3 c}{\partial y^3}\Big|_{i,j} + O(\Delta y^4)$$

→ Backward Expansion

$$c_{i,j-1} = c_{i,j} - \Delta y \frac{\partial c}{\partial y}\Big|_{i,j} + \frac{\Delta y^2}{2!}\frac{\partial^2 c}{\partial y^2}\Big|_{i,j} - \frac{\Delta y^3}{3!}\frac{\partial^3 c}{\partial y^3}\Big|_{i,j} + O(\Delta y^4)$$

→ Sum

$$c_{i,j+1} + c_{i,j-1} = 2c_{i,j} + \Delta y^2 \frac{\partial^2 c}{\partial y^2}\Big|_{i,j} + O(\Delta y^4)$$

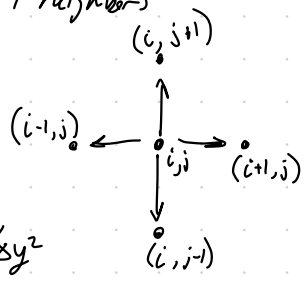
$$\frac{\partial^2 c}{\partial y^2}\Big|_{i,j} \approx \frac{c_{i,j+1} - 2c_{i,j} + c_{i,j-1}}{\Delta y^2} + O(\Delta y^4)$$

→ Combine for full discretization

$$\frac{\partial c_{i,j}}{\partial t} = \alpha\left(\frac{c_{i+1,j} - 2c_{i,j} + c_{i-1,j}}{\Delta x^2} + \frac{c_{i,j+1} - 2c_{i,j} + c_{i,j-1}}{\Delta y^2}\right)$$

→ Couples (i,j) to itself & 4 neighbors

Cell	Coefficient
$c(i+1,j)$	$\alpha/\Delta x^2$
$c(i-1,j)$	$\alpha/\Delta x^2$
$c(i,j+1)$	$\alpha/\Delta y^2$
$c(i,j-1)$	$\alpha/\Delta y^2$
$c(i,j)$	$-2\alpha/\Delta x^2 - 2\alpha/\Delta y^2$



Q1.1 Part A Boundry Conditions for Ghost Cells

→ Inlet & Outlet

for a Dirichlet Condition the face value is the simple linear avg of ghost cell & 1st Real Cell

$$C_{face} = \frac{C_{ghost} + C_{real}}{2} = C_{bc}$$

$$C_g = 2C_{bc} - C_{real}$$

Inlet: left Boundry $i=0$ $(0.86 < y_j < 0.94)$

$$C_{j,0} = 2C_{in} - C_{j,1} = 2(1) - C_{j,1} = 2 - C_{j,1}$$

Outlet: Right Boundry $i = N_x+1$ $(0.06 < y_j < 0.14)$

$$C_{j,N_x+1} = 2C_{out} - C_{j,N_x} = 2(0) - C_{j,N_x} = -C_{j,N_x}$$

→ Solid Walls → Neumann Zero flux Ghost Cell

at non in/outlet segments $\partial c/\partial n = 0$ so normal gradient $\left.\frac{\partial c}{\partial n}\right|_{face} = 0 \Rightarrow \frac{C_{real} - C_{ghost}}{\Delta x} = 0$

so $C_{real} = C_{ghost}$

→ Neumann to 4 edges

$C_{j,0} = C_{j,1}$ Left

$C_{j,N_x+1} = C_{j,N_x}$ Right

$C_{0,i} = C_{1,i}$ Bottom

$C_{N_y+1,i} = C_{N_y,i}$ Top

→ Summary

Boundry	Segment	Condition	Ghost Cell
Left $(i=0)$	Inlet	Dirichlet $c=1$	$C_{j,0} = 2 - C_{j,1}$
Left $(i=0)$	Left Wall	Neumann $\partial c/\partial x = 0$	$C_{j,0} = C_{j,1}$
Right $(i=N_x+1)$	outlet	Dirichlet $c=0$	$C_{j,N_x+1} = -C_{j,N_x}$
Right $(i=N_x+1)$	Right wall	Neumann $\partial c/\partial x = 0$	$C_{j,N_x+1} = C_{j,N_x}$
Bottom $(j=0)$	wall	Neumann $\partial c/\partial y = 0$	$C_{0,i} = C_{1,i}$
Top $(j=1)$	wall	Neumann $\partial c/\partial y = 0$	$C_{N_y+1,i} = C_{N_y,i}$